



GIRRAWEE HIGH SCHOOL

YEAR 11 - TASK 1

2004

MATHEMATICS

2 UNIT

Time allowed – 90 minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a *new* sheet of paper.

Yr 11 Mathematics - Task 1 (2004)

Question 1 (16 marks)

- (a) Find $\sqrt{\frac{3}{7^2}} + 4(5)^3$ correct to 2 decimal places. 2
- (b) Evaluate $\sqrt{\frac{83.1}{0.2 + 5.4 \div 1.3}}$ correct to 3 significant figures. 2
- (c) Write in scientific notation.
(i) 364 000 2
(ii) 0.000912 2
- (d) Simplify $2\sqrt{12}$. 2
- (e) If $\sqrt{800} = x\sqrt{2}$, find the value of x . 2
- (f) Write $64^{-\frac{2}{3}}$ as a rational number. 2
- (g) Find $(5.7 \times 10^{13}) \times (4.9 \times 10^{22})$. Write your answer in scientific notation. 2

Question 2 (16 marks)

- (a) Expand and simplify
- (i) $4a(a - 2b) - 2a(b - 3a)$ 2
- (ii) $(3y - 1)(3y + 1)$ 2
- (iii) $(x - 5)(x^2 + 5x + 25)$ 2
- (iv) $(5t + 3)^2$ 2
- (v) $(2x - 3)(x + 8)$ 2
- (b) If $d = \frac{c}{\sqrt{a^2 + b^2}}$. Find the exact value of d with a rational denominator if
 $a = 2, b = -1, c = 3,$ 2
- (c) If $A = \frac{1}{2}r^2\theta$. Find the value of θ , when $A = 12$ and $r = 4$. 2
- (d) If $V = \pi r^2 h$. Find the *exact* value of r when $V = 9$ and $h = 16$. 2

Question 3 (14 marks)

- (a) Factorise.
- (i) $m^2 + 14m - 32$ 2
- (ii) $x^3 + x^2 + 3x + 3$ 2
- (iii) $b^3 - 8$ 2
- (iv) $3y^2 + 5y - 2$ 3
- (v) $z^2 - (1 + w)^2$ 3
- (b) Complete the square on, $x^2 - 18x + \square$? 2

Question 4 (23 marks)

- (a) Simplify.
- (i) $\sqrt{27} - \sqrt{12}$ 2
- (ii) $\sqrt{32} - 3\sqrt{2} + 2\sqrt{50}$ 2
- (iii) $\frac{\sqrt{2}}{\sqrt{8}}$ 2
- (b) Expand and simplify.
- (i) $2\sqrt{3}(2 - \sqrt{12})$ 2
- (ii) $(2\sqrt{2} - \sqrt{3})^2$ 3
- (c) Express with a rational denominator.
- (i) $\frac{5}{\sqrt{2}}$ 2
- (ii) $\frac{\sqrt{3}}{\sqrt{2} - 7}$ 3
- (iii) $\frac{\sqrt{3}}{\sqrt{5} + 2\sqrt{6}}$ 3
- (d) Express as a single fraction with rational denominator.
- $\frac{\sqrt{2} - \sqrt{7}}{\sqrt{2} + \sqrt{3}} \times \frac{2}{\sqrt{2} - \sqrt{3}}$ 4

Question 5 (20 marks)

- (a) Simplify
- (i) $\frac{4a}{7} \times \frac{21}{5a^2}$ 2
- (ii) $\frac{5}{x^2 - 1} \div \frac{2}{x - 1}$ 2
- (iii) $\frac{3}{b + 2} \times \frac{b^2 + 2b}{6a - 3}$ 3
- (iv) $\frac{3ab^2}{5xy} + \frac{12ab - 6a}{x^2y + 2xy^2}$ 3
- (b) Simplify
- (i) $\frac{a + 4}{3} - \frac{a}{4}$ 2
- (ii) $\frac{1}{a^2 + 2a + 1} + \frac{1}{a + 1}$ 2
- (iii) $x - \frac{x^2}{x + 2}$ 3
- (iv) $\frac{1}{x - 1} - \frac{2}{x}$ 3

Question 6 (15 marks)

- (a) Simplify. $\frac{2xy + 2x - 6 - 6y}{4x^2 - 16x + 12}$ 4
- (b) Simplify. $\frac{2p^2 - 3p - 2}{8p^3 + 1}$ 4
- (c) Write $0.\dot{3}0\dot{4}$ as a rational number. 3
- (d) Write $1.6\dot{2}\dot{7}$ as a rational number. 4

Solutions.

Yr 11 - Mathematics

TASK 1 (2004)

$$\textcircled{Q1} \text{ a) } = 500.2474... \\ = 500.25 \text{ (2 dec. pl.)}$$

$$\text{b) } = 4.368818... \\ = 4.37 \text{ (3 sign figs.)}$$

$$\text{c) (i) } 3.64 \times 10^5$$

$$\text{(ii) } 9.12 \times 10^{-4}$$

$$\text{d) } = 2 \times 2\sqrt{3} \\ = 4\sqrt{3}$$

$$\text{e) } \sqrt{800} = \sqrt{400 \times 2} \\ = 20\sqrt{2} \\ \therefore x = 20$$

$$\text{f) } 64^{-\frac{2}{3}} = \frac{1}{\sqrt[3]{64^2}} = \frac{1}{16}$$

$$\text{g) } 2.793 \times 10^{36}$$

$$\textcircled{Q2} \text{ a) } 4a(a-2b) - 2a(b-3a) \\ = 4a^2 - 8ab - 2ab + 6a^2 \\ = 10a^2 - 10ab$$

$$\text{(ii) } (3y-1)(3y+1) \\ = 9y^2 - 1$$

$$\text{(iii) } (x-5)(x^2+5x+25) \\ = x^3 - 5^3 \\ = x^3 - 125$$

$$\text{(iv) } (5t+3)^2 = 25t^2 + 30t + 9$$

$$\text{(v) } (2x-3)(x+8) = 2x^2 + 16x - 3x - 24 \\ = 2x^2 + 13x - 24$$

$$\text{b) } d = \frac{3}{\sqrt{2^2 + (-1)^2}} = \frac{3}{\sqrt{5} \times \sqrt{5}} \\ = \frac{3\sqrt{5}}{5}$$

$$\text{c) } 12 = \frac{1}{2}(4)^2\theta \\ 24 = 16\theta \\ \theta = \frac{24}{16} \\ \theta = 1.5 \text{ or } 1\frac{1}{2}$$

$$\text{d) } 9 = \pi r^2 / 16 \\ r^2 = \frac{9}{\pi(16)} \\ r = \pm \sqrt{\frac{9}{16\pi}} = \pm \frac{3}{4\sqrt{\pi}}$$

$$\textcircled{Q3} \text{ a) (i) } m^2 + 14m - 32 \\ = (m+16)(m-2)$$

$$\text{(ii) } x^3 + x^2 + 3x + 3 \\ = x^2(x+1) + 3(x+1) \\ = (x+1)(x^2+3)$$

$$\textcircled{Q3} \text{ a) (iii) } b^3 - 8 \\ \text{Factor: } = (b-2)(b^2+2b+4)$$

$$\text{(iv) } 3y^2 + 5y - 2 \\ \text{or } (3y-1)(y+2) = 3y^2 + 6y - y - 2 \\ = 3y^2 + 5y - 2 \\ = (y+2)(3y-1)$$

$$\text{(v) } z^2 - (1+w)^2 \\ = (z-(1+w))(z+(1+w)) \\ = (z-1-w)(z+1+w)$$

$$\text{b) } x^2 + 18x + \square \\ \square = \left(\frac{-18}{2}\right)^2 = 81$$

$$\textcircled{Q4} \text{ a) (i) } \sqrt{27} - \sqrt{12} \\ = 3\sqrt{3} - 2\sqrt{3} \\ = \sqrt{3}$$

$$\text{(ii) } \sqrt{32} - 3\sqrt{2} + 2\sqrt{50} \\ = 4\sqrt{2} - 3\sqrt{2} + 2 \times 5\sqrt{2} \\ = \sqrt{2} + 10\sqrt{2} \\ = 11\sqrt{2}$$

$$\text{(iii) } \frac{\sqrt{2}}{\sqrt{8}} = \frac{\sqrt{2}}{2\sqrt{2}} = \frac{1}{2}$$

$$\text{b) (i) } 2\sqrt{3}(2-\sqrt{12}) \\ = 4\sqrt{3} - 2\sqrt{36} \\ = 4\sqrt{3} - 12$$

$$\text{(ii) } (2\sqrt{2} - \sqrt{3})^2 \\ = 8 - 4\sqrt{6} + 3 \\ = 11 - 4\sqrt{6}$$

$$\text{c) (i) } \frac{5}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{5\sqrt{2}}{2}$$

$$\text{(ii) } \frac{\sqrt{3}}{\sqrt{2}-7} \times \frac{\sqrt{2}+7}{\sqrt{2}+7} = \frac{\sqrt{6}+7\sqrt{3}}{2-49} \\ = \frac{\sqrt{6}+7\sqrt{3}}{-47}$$

$$\text{(iii) } \frac{\sqrt{3}}{\sqrt{5}+2\sqrt{6}} \times \frac{\sqrt{5}-2\sqrt{6}}{\sqrt{5}-2\sqrt{6}} \\ = \frac{\sqrt{15}-2\sqrt{18}}{5-24} \\ = \frac{\sqrt{15}-6\sqrt{2}}{-19}$$

$$\text{d) } \frac{\sqrt{2}-\sqrt{7}}{\sqrt{2}+\sqrt{3}} \times \frac{2}{\sqrt{2}-\sqrt{3}} \\ = \frac{2\sqrt{2}-2\sqrt{7}}{2-3} \\ = \frac{2\sqrt{2}-2\sqrt{7}}{-1} \text{ or } 2\sqrt{7}-2\sqrt{2}$$

Q5 a) i) $\frac{4x}{7} \times \frac{2x^3}{5a^2} = \frac{12}{5a}$

ii) $\frac{5}{x^2-1} \cdot \frac{2}{x-1}$
 $= \frac{5}{(x+1)(x-1)} \times \frac{2}{x-1}$
 $= \frac{5}{2(x+1)} \quad \text{or} \quad \frac{5}{2x+2}$

iii) $\frac{3}{b+2} \times \frac{b^2+2b}{6a-3}$
 $= \frac{3}{b+2} \times \frac{b(b+2)}{3(2a-1)}$
 $= \frac{b}{2a-1}$

iv) $\frac{3ab^2}{5xy} \cdot \frac{12ab-6a}{x^2y+2xy^2}$
 $= \frac{3ab^2}{5xy} \times \frac{xy(x+2y)}{xy(x+2y)}$
 $= \frac{b^2(x+2y)}{10(2b-1)}$
 or $= \frac{b^2x+2yb^2}{20b-10}$

b) i) $\frac{a+4}{3} - \frac{a}{4}$
 $= \frac{4(a+4) - 3a}{12}$
 $= \frac{4a+16-3a}{12}$
 $= \frac{a+16}{12}$

ii) $\frac{1}{a^2+2a+1} + \frac{1}{a+1}$
 $= \frac{1}{(a+1)^2} + \frac{1}{a+1}$
 $= \frac{1 + (a+1)}{(a+1)^2}$
 $= \frac{a+2}{(a+1)^2}$

iii) $\frac{x-x^2}{x+2} = \frac{x(x+2)}{x+2} - \frac{x^2}{x+2}$
 $= \frac{x^2+2x-x^2}{x+2}$
 $= \frac{2x}{x+2}$

iv) $\frac{1}{x-1} - \frac{2}{x} = \frac{x-2(x-1)}{x(x-1)}$
 $= \frac{x-2x+2}{x(x-1)}$
 $= \frac{2-x}{x(x-1)}$

Q6 a) $\frac{2xy+2x-6-6y}{4x^2-16x+12}$
 $= \frac{2x(y+1)-6(y+1)}{4(x^2-4x+3)}$
 $= \frac{(y+1)(2x-6)}{4(x-3)(x-1)}$
 $= \frac{2(y+1)(x-3)}{2 \cdot 4(x-1)(x-3)}$
 $= \frac{y+1}{2x-2}$

Q6 cont.

b) $\frac{2p^2-3p-2}{8p^3+1} \times \frac{p}{p-2}$
 $= \frac{(2p+1)(p-2)}{(2p+1)(4p^2-2p+1)}$
 $= \frac{p-2}{4p^2-2p+1}$

c) $x = 0.304$
 $x = 0.304304...$
 $1000x = 304.304304...$
 $1000x - x = 304.304... - 0.304...$
 $999x = 304$
 $x = \frac{304}{999}$

d) $x = 1.627$
 $x = 1.62727...$
 $10x = 16.2727...$
 $1000x = 1627.2727...$
 $1000x - 10x = 1627.2727... - 16.2727...$
 $990x = 1627 - 16$
 $990x = 1611$
 $x = \frac{1611}{990}$
 $x = 1\frac{61}{110}$